Board 2 - Emmanuel Garcia: Grip Actuation Node

Bidirectional H-Bridge
Manufacturer
Part #: MD135

DC Motor & Lead-Screw

M+

M-

PWM

DIR

Return current

Current Sense & Overload
Texas Instruments TLV9002

Low-side shunt

Op-amp comparator

Overload alert

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RC0 PWM

RC1 GPIO

RA1 ADC

Digital I/O (UART + GPIO)

RD2 RX | RD3 TX | RB0 ESTOP | RE1 ALERT

1 2 3 4 5 6 7 8

digital | analog | GND

Connector 1 (to Board 1 hub)

8-pin ribbon cable

Board 3 - Jared: Tactile Force Sensing Node

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PWM RA2 → Status LED

GPIO RC4 ← Tare Button

ADC RA0 ← Stage 2: Active Low-Pass Filter

Load Cell
Manufacturer
Part #

Stage 1: Instrumentation Amp
Manufacturer
Part #

Stage 2: Active Low-Pass Filter
Manufacturer
Part #

Digital I/O (UART + GPIO)

RB1 RX | RB2 TX | RB0 ESTOP | RB3 ALERT

1 2 3 4 5 6 7 8

digital | analog | GND

Connector 1 (to Board 1 hub)

8-pin ribbon cable

Board 1 - Sakiya: Bus Master & UI Node (hub)

Connector 1 (to Board 2)

digital | analog | GND

1 2 3 4 5 6 7 8

RB0 TX2 | RB1 RX2 | RD4 ESTOP | RC4 ALERT

Connector 2 (to Board 3)

digital | analog | GND

1 2 3 4 5 6 7 8

RA6 TX3 | RA7 RX3 | RD5 ESTOP | RC5 ALERT

Digital I/O (UART + GPIO)

DC Supply → V+ Power

E-Stop Button (SW0) → RD0 GPIO

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PWM RA2 → Status LED

UART1 RF0 (TX) → Host PC (USB debug UART)

UART1 RF1 (RX) ←

Digital I/O (UART + GPIO)

RD2 TX4 | RD3 RX4 | RD6 ESTOP | RC6 ALERT

RA4 TX5 | RA5 RX5 | RD7 ESTOP | RC7 ALERT

1 2 3 4 5 6 7 8

digital | analog | GND

Connector 3 (to Board 4)

Connector 4 (to Board 5)

8-pin ribbon cable

8-pin ribbon cable

Ribbon cable pinout
(same on all four cables)

1: UART, hub → board
2: UART, board → hub
3: E-STOP, hub → board
4: ALERT, board → hub
5-7: not used (spare)
8: GND

Board 4 - Joseph Z: Object Proximity Node

Connector 1 (to Board 1 hub)

digital | analog | GND

1 2 3 4 5 6 7 8

RB1 RX | RB2 TX | RB0 ESTOP | RB3 ALERT

Digital I/O (UART + GPIO)

IR Photodiode Pair
Manufacturer
Part #

Op Amp Band-Pass Filter
Manufacturer
Part #

RA0 ADC

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Board 5 - Aakash Kumar: Accelerometer Slip Detection Node

Connector 1 (to Board 1 hub)

digital | analog | GND

1 2 3 4 5 6 7 8

RB1 RX | RB2 TX | RB0 ESTOP | RB3 ALERT

Digital I/O (UART + GPIO)

Signal Conditioning Array
6x Texas Instruments TLV9062

ADC RA0 ← Peak detection ← High-pass filter ← X

ADC RA6 ← Peak detection ← High-pass filter ← Y

ADC RA5 ← Peak detection ← High-pass filter ← Z

3-Axis Accelerometer
Analog Devices
Part #: ADXL335BCPZ

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